

Shoulder Injuries

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Christoph Lutter climbing in Mill Creek Canyon, Moab, Utah, USA,
Photo Michael Simon

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Table 9.1 Current distribution of shoulder injuries in rock climbing (Schöffl et al. [2])

Shoulder injuries 2017–2018 (<i>n</i> = 154)	<i>n</i>	%	Shoulder injuries 2009–2012 (<i>n</i> = 157)	<i>n</i>	%
SLAP lesion	37	29.8	SLAP lesion	51	32.5
Impingement syndrome	34	27.4	Impingement syndrome	40	25.5
Shoulder dislocation, Bankart lesion	22	17.7	Shoulder strain	17	10.8
Shoulder strain	16	12.9	Shoulder dislocation, Bankart lesion	16	10.2
Rotator cuff tear	12	9.7	Supraspinatus tendonitis	7	4.5
Acromioclavicular joint injury	12	9.7	Functional instability	7	4.5
Tendinopathy of the long head of the biceps tendon	6	4.8	Tendinopathy of the long head of the biceps tendon	5	3.2
Functional instability	5	4.0	Rupture of the long head of the biceps tendon	5	3.2
Pulley lesion	5	4.0	Rotator cuff tear	5	3.2
Rupture of the long head of the biceps tendon	2	1.6	Acromioclavicular joint injury	3	1.9
Others	2	1.6	Pulley lesion	1	0.6
Supraspinatus tendonitis	1	0.8			

9.1 Overview

Shoulder injuries are the second most common pathologies in rock climbing after finger injuries, whereby numbers have increased in the last couple of years [1].

Within our own patient cohort, we found an increase of the incidence of shoulder injuries from 5% (1998–2001) to 20.2% from 2017 to 2018 [2]. SLAP lesions (superior labrum anterior to posterior) play an essential role as they are the fifth most common diagnosis (5.8%) out of all climbing-related pathologies [2]. Table 9.1 shows the current distribution of shoulder injuries among rock climbers from our own patient cohort. In the following sections, we discuss the specifics of the essential climbing-related shoulder injuries as well as their diagnostics and therapy.

9.2 Impingement Syndrome

Impingement symptoms are the second most common shoulder pathology in rock climbers [3]. Regarding etiology, we have to distinguish between the primary and secondary forms. Primary impingement syndrome is caused by

structural changes that mechanically narrow the subacromial space. These changes include bony narrowing from the cranial side (outlet impingement), which is the case in acromial Types II–III according to Bigliani and Morrison, structural variations of the coracoid, and bony malposition after a fracture of the greater tubercle. An increase in the volume of the subacromial tissue (e.g., calcific tendonitis, subacromial bursitis) may also lead to a primary impingement (non-outlet impingement). Secondary impingement is due to a functional disturbance of the centering of the humeral head triggered by muscular imbalance. This leads to an abnormal displacement of the center of rotation in elevation and results in soft tissue entrapment [4, 5]. Another relatively rare form of functional impingement syndrome in climbers is coracoid impingement, which is caused by a hypertrophied subscapularis tendon and muscle (Fig. 9.1) following intense sport-specific training [6]. Finally, internal impingement can also be the reason for impingement symptoms in athletes. In this case, the mechanism is a repetitive squeezing of the supra- and infraspinatus tendons against the glenoid in extreme external rotation and simultaneous abduction (posterosuperior) or rather internal rotation and adduction (anterosuperior) [7, 8]. Additionally, symptoms can be aggravated by

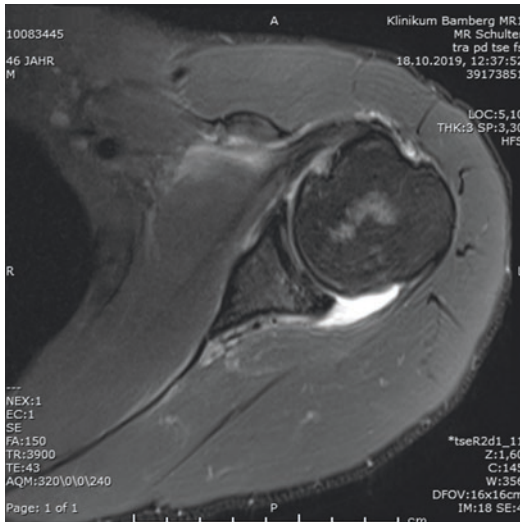


Fig. 9.1 MRI of a shoulder with coracoid impingement



Fig. 9.2 Postural deformity, the so-called “climber’s back”

degenerative alterations of the acromioclavicular joint as well as a postural deformity, the so-called “climber’s back” (Fig. 9.2) (see also Chap. 12 Spine). Shortened pectoral muscles and pro-

nounced rear shoulder girdle muscles result in an increased thoracic kyphosis, leading to an amplified internal rotation of the shoulders, which ultimately enhances preexisting impingement symptoms [9].

Clinically, climbers complain of persistent pain without a preceding trauma, especially when elevating the arm (in particular between 70° and 120°; painful arc), during forced movement above the head, and also during rest when lying on the affected side [4, 5].

The diagnosis is made mainly by a thorough clinical examination (sensitivity up to 90%; e.g., Hawkins test, Neer’s sign, Jobe test, painful arc), ultrasound, and plain X-rays. In case of persistent complaints, an MRI is mandatory to evaluate the condition of the rotator cuff, the bursa, and the labrum-ligament complex. Aspects such as potential tears of the rotator cuff, the degree of atrophy (according to Thomazeau), and fatty degeneration (according to Goutallier) of the muscle bellies of the rotator cuff can be assessed [5]. In general, literature recommends conservative therapy (physical therapy and nonsteroidal anti-inflammatory drugs). In case of sustained complaints, subacromial infiltrations with local anesthetics and corticosteroids (3–4 weeks after onset of symptoms, at the earliest, and no more often than two to three times, no infiltration into the tendon) under ongoing physical therapy are recommended. Corticosteroid injections may be restricted by the World Anti-Doping Agency (WADA) and may need specific documentation (a definitive statement is not possible here, as the regulations are constantly changed and revised.) In the case of an accompanying calcific tendinitis, shockwave therapy shows good results. Overall, more than 70% of all impingement symptoms can be treated successfully through conservative therapy. If conservative therapy fails for 3 months or longer or in the presence of other comorbidities (e.g., rotator cuff tear, SLAP lesion, etc.), surgical therapy is indicated. Hereby, an arthroscopic, subacromial decompression is performed. Besides a bursectomy, this also includes the removal of the anterior and lateral portions of the undersurface of the acromion (5–8 mm). If necessary, the removal of inferior

acromial osteophytes and the lateral end of the clavicle without a total resection of the acromioclavicular joint can be performed as part of the same arthroscopic surgery (coplaning) [5].

Impingement Syndrome

- **History:** Congenital deformity, muscular imbalance
- **Onset:** Chronic
- **Clinical:** Pain without trauma, especially on forced movement above the head and when lying on the affected side
- **Diagnostic:** Medical history, clinical finding, sonography, if necessary MRI
- **Therapy:** Mostly conservative
- **Outcome:** Good

9.3 SLAP Lesions

SLAP lesions (superior labrum anterior to posterior) are the most common shoulder injuries in rock climbers (29.8%) [2]. “Shoulder moves” are particularly stressful for the insertion of the long biceps tendon (Schöffl et al. 2011). During these moves, a subsequent internal rotation and anterior tilt of the shoulder under load leads to increased tensile strain on the pulley system and the labro-bicipital complex (LBC). Nevertheless, SLAP injuries are usually caused by repetitive microtrauma rather than one single incident. Thus, it is not surprising that most of the affected climbers are older than 35 years with many years of climbing experience [11, 12]. Patients presenting with acute injuries usually describe a fall onto an outstretched arm or a sudden, unexpected pull on the arm (e.g., a climber trying to hold on while the feet are slipping).

Clinically, climbers complain of a sudden or gradual deterioration of shoulder function and concomitant pain. In the majority of cases, the dominant arm is concerned. The term “dead arm syndrome” specifies the inability to execute sport-specific movements at pre-injury velocity [13]. Other symptoms include sensations of intermittent clicking or popping during the cocking phase and anterior shoulder pain [14]. A high number of clinical tests (e.g., O’Brien test, crank



Fig. 9.3 Climbing-specific SLAP tests (Photo: Michael Simon)

test, dynamic labral shear test) have been described for the detection of SLAP lesions, but no gold standard with regard to sensitivity or specificity has been established yet. Thus, two climbing-specific SLAP provocation tests (Fig. 9.3) are recommended, as they simulate the sport-specific strain on the labro-bicipital complex. To confirm the diagnosis, magnetic resonance arthrography (MRA) is mandatory [15] (Fig. 9.4). According to current literature, SLAP lesions with structural damage, which is more than a fraying (Grade II or higher, according to Snyder) to the biceps tendon anchor or the labrum, should be treated surgically in athletes with a primary tenodesis of the long biceps tendon. SLAP I lesions can be treated conservatively. Our standard surgical approach in overhead sports athletes is a tenodesis of the long biceps tendon at the top of the bicipital groove with a composite tenodesis screw [16, 17]. Against the background of repetitive microtrauma, a solid tenodesis of the long head of the biceps tendon and thus the deactivation of the pain generator in the area of the LBC are reasonable [10, 18]. In

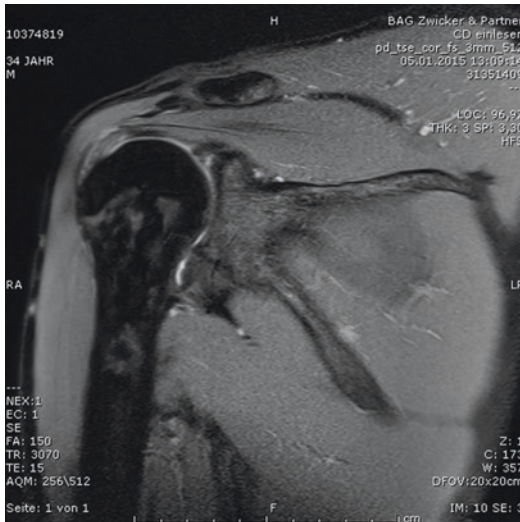


Fig. 9.4 MRI of a shoulder with a SLAP II lesion (Snyder classification)

addition to the tenodesis of the long head of the biceps tendon, the superior labrum is refixed. Although some studies have shown a good outcome in 75–97% of nonathletes treated with an orthotopic refixation of the long biceps tendon to the LBC, other studies focusing on the outcome in overhead athletes found only 22–75% satisfactory results. Thus, for overhead athletes, a primary tenodesis of the long head of the biceps tendon is the therapy of choice [18, 19]. Nevertheless MRI evaluation on asymptomatic climbers recently showed a high prevalence of labral and other pathologies, thus the clinical presentation is most important to finally decide who needs a surgical revision [20].

This is also highlighted by the findings of a follow-up examination of 30 climbers, who were treated with a primary tenodesis of the long head of the biceps tendon and a refixation of the superior labrum, if necessary. In this cohort, we found good results with an average Constant-Murley score of 89.6 points. All patients had started climbing again after 12 months following surgery, and 67% had already regained their pre-injury climbing level [16].

SLAP Lesions

- **History:** Repetitive microtrauma, acute trauma
- **Onset:** Chronic and acute

- **Clinical:** Sudden or gradual deterioration of shoulder function and concomitant pain
- **Diagnostic:** Medical history, clinical finding, MRA
- **Therapy:** Surgical therapy in case of SLAP lesions Grade II (Snyder) or higher
- **Outcome:** Good

9.4 Pathologies of the Rotator Cuff

A growing number of aging athletes is still climbing at a high difficulty level [12]. As such, the number of degenerative pathologies, such as tears of the rotator cuff (consisting of supraspinatus, infraspinatus, and teres minor muscles), is also increasing. Additionally, injuries to the rotator cuff can also be caused by acute trauma (e.g., shoulder dislocation, a direct fall on the arm in abduction, or a sudden unexpected pull on the arm). The most common type of injury is a rupture of the tendon of the supraspinatus muscle. Clinically, climbers complain of a sudden or gradual onset of pain during rest but mainly during abduction over more than 90° in combination with loss of strength [21]. Clinical symptoms often depend on the location and size of the respective tear. Frequently, there is an inverse correlation between tear size and pain intensity. In practice, partial thickness rotator cuff tears on the bursal surface can be extremely painful, whereas massive tears are mainly associated with a loss of strength and a feeling of instability. The diagnosis is made by a thorough medical history, a clinical examination, ultrasound, and MRI.

Isometric function tests enable the physician to examine the different muscles of the rotator cuff regarding strength and pain (e.g., *supraspinatus muscle*—Jobe test, starter test, external rotation lag sign, drop arm sign; *subscapularis muscle*—lift-off test, internal rotation lag sign, belly press test). The condition of the rotator cuff, the bursa, and the labrum-ligament complex can be investigated with sonography and MRI. The tear size as well as the degree of atrophy (according to Thomazeau) and fatty degeneration (according to Goutallier) of the muscle bellies of the rotator cuff can be assessed [22]

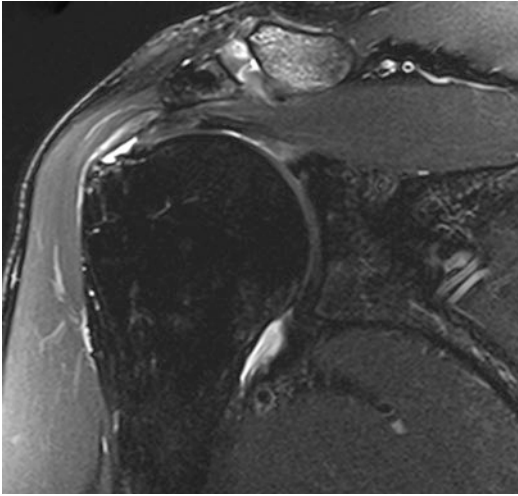


Fig. 9.5 MRI of a shoulder with a complete rotator cuff tear (m. supraspinatus)

(Fig. 9.5). While small partial thickness tears can be treated conservatively, full thickness tears and severe partial thickness tears in athletes should be treated surgically [11]. According to current literature, arthroscopic surgical techniques using composite suture anchors are the standard procedure. A retrospective study by Simon et al. analyzed the general (Constant-Murley score) and sport-specific (change in International Climbing and Mountaineering Federation [UIAA] grade) outcome after the surgical repair of rotator cuff injuries in 12 rock climbers and found a good functional outcome. The mean Constant-Murley score was 92 points. Furthermore, all patients had restarted climbing within 12 months after surgery [23].

Pathologies of the Rotator Cuff

- **History:** Chronic degenerative changes, acute trauma
- **Onset:** Chronic and acute
- **Clinical:** Pain and loss of strength, especially at abduction over 90°
- **Diagnostic:** Medical history, clinical finding, sonography, MRI
- **Therapy:** Surgical therapy in case of massive full thickness tears or severe partial tears
- **Outcome:** Good

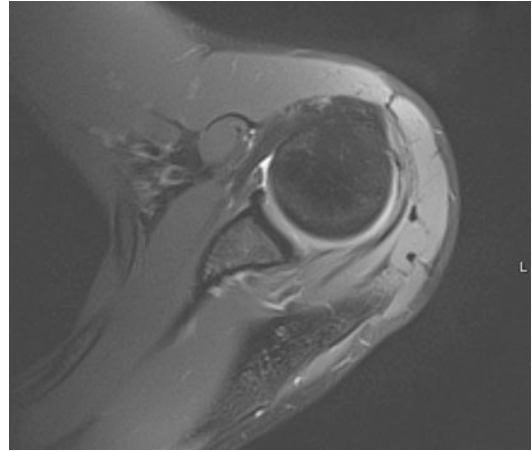


Fig. 9.6 MRI of a shoulder with a Bankart lesion following a traumatic shoulder dislocation

9.5 Shoulder Dislocation

The shoulder joint has the biggest range of motion of all human joints, making it particularly vulnerable to dislocation. Approximately 75% of all primary shoulder dislocations happen during sports, with the shoulder joint in an abduction and externally rotated position.

Unidirectional anterior dislocations make up the majority of cases (95%). As a result of the direction of the dislocation, the particular trauma mechanism, and the age of the patient, injuries of the labrum-ligament complex, the rotator cuff, and bony injuries of the humeral head (Hill-Sachs lesion) or the glenoid (Bankart fracture) may occur. In most cases, these injuries lead to instability with an increased risk of repeated shoulder dislocations, especially during overhead activities. Following a shoulder dislocation, patients complain of load-dependent pain and a pronounced feeling of instability. The diagnosis is made by a thorough medical history and a clinical examination of both shoulders (apprehension test, relocation test, load and shift test, sulcus sign), including neurological tests to exclude injuries to the brachial plexus. Furthermore, an MRI (potentially with an intra-articular contrast agent—MRA) is indispensable [24] (Fig. 9.6). Besides traumatic lesions of the labrum due to

traumatic shoulder dislocations (Polar Type I), we also occasionally find pathologies in climbers on the basis of chronic microtrauma (Polar Type III), which lead to functional instability. These cases are accompanied by laxity of the joint capsule and a dysfunctional locomotion pattern, both leading to recurrent shoulder subluxations, whereby structural pathologies of the labrum-ligament complex are not usually found [11].

The climbers' complaints are often associated with specific climbing-related movements, such as passively hanging from one arm without muscular stabilization. These so-called dead hangs can cause mechanoreceptor damage and laxity of the joint capsule. Besides a mechanic insufficiency, a neuromuscular insufficiency of the joint stabilization may also occur [11, 25, 26].

In most cases, the standard therapy for structural lesions of the labrum-ligament complex or fractures of the glenoid in athletes following a shoulder dislocation is an arthroscopic surgery [11]. In this way, the development of a chronic instability can be avoided.

In a recent study, ten climbers from our own patient collective were treated with an arthroscopic refixation of the labrum-ligament complex following a traumatic shoulder dislocation. In the follow-up, we found a very favorable functional and sport-specific outcome. All climbers achieved their pre-injury performance level, no recurrent dislocation had occurred, and the mean Constant-Murley score was 97 points [27].

The therapy for climbers with a functional instability of the shoulder joint (Polar Type III) is always conservative by means of stabilization training and a modification in climbing technique [11, 25, 26].

Shoulder Dislocation

- **History:** Trauma, functional
- **Onset:** Acute, acquired
- **Clinical:** Pain and instability
- **Diagnostic:** Medical history, clinical finding, sonography, MRI (potentially with an intra-articular contrast agent—MRA)

- **Therapy:** Surgical therapy in case of structural pathologies of the labrum-ligament complex
- **Outcome:** Good

9.6 Rupture of the Long Head of the Biceps Tendon

Since climbing can often be performed at a high to very high level into old age, degenerative ruptures of the long head of the biceps tendon (LHBT) and its guidance through the so-called pulley are not uncommon due to the biomechanical load. There is a clear increase in the incidence from the age of 45 onward [12]. Patients usually report preexisting symptoms and often minor trauma, which ultimately leads to a final rupture of the LHBT. A sudden event with stabbing, whiplike pain is indicated in the medical history, after which the impressive “Popeye sign,” a full distalization of the muscle belly, is clinically evident (Fig. 9.7). There is usually a tenderness of the intertubercular sulcus, but the loss of strength is minimal. The diagnosis is carried out by sonography and MRI, in addition to the clinical examination [28].



Fig. 9.7 “Popeye sign” following a rupture of the distal biceps tendon

A conservative therapy of LHBT ruptures results in a force reduction of 8–21% for flexion and supination in the elbow and a persistent cosmetic deformity (“Popeye sign”). After surgical refixation, this strength deficit is only 0–10%. Since climbers usually do not accept a strength deficit, surgical therapy (tenodesis) is recommended [11].

All athletes treated in this way regained their previous sport-specific climbing level. In a retrospective study, an average Constant-Murley score of 97.8 points and a self-evaluated biceps function of 97.3% were determined [29].

Rupture of the Long Head of Biceps Tendon

- **History:** Chronic, degenerative changes
- **Onset:** Acute to chronic
- **Clinical:** Pain, Popeye sign, only minimal loss of strength
- **Diagnostic:** Medical history, clinical finding, sonography, MRI
- **Therapy:** Surgical
- **Outcome:** Good

9.7 Thrombosis of the Subclavian Vein (Paget-Schroetter Syndrome)

Deep vein thrombosis of the upper extremities (Paget-Schroetter syndrome) is rare overall (2 per 100,000 patients per year) but, nevertheless, represents one of the most common entities of vascular pathologies in athletes. Especially because of the growing popularity of climbing and bouldering, this diagnosis, which occurs primarily in sports with particularly high strain on the upper extremities, is expected to increase in the future. The probable cause is a mechanical external compression of the subclavian or the axillary vein in its course between the clavicle and the first rib through the musculoligamentous structures (e.g., scalenus anterior, muscle, subclavius muscle, costoclavicular ligament). Compression appears especially during abduction or particularly strenuous arm movements, which often occur during climbing.

Repetitive training protocols might induce intimal lesions and thus cause fibrosis. The resulting increased scar tissue leads to a progressive stenosis. Under load, this can no longer be compensated by collaterals, and even a small thrombus can cause the lumen to become completely blocked. Consequently, the restricted blood flow can no longer be compensated, at which point symptoms appear [30, 31]. Patients report non-specific shoulder pain and feelings of tension in the shoulder and neck area. In addition, there is often a feeling of pressure in the entire arm, a change of the skin color in the sense of a livid discoloration up to the formation of ecchymoses, and a tendency of swelling during or after climbing. In the case of a subacute course, the development of collateral circulation with increased venous drawing, especially following exercise, can be found in the shoulder area (Fig. 9.8). Since the initial symptoms are often unspecific, it is important for the treating physician to be familiar with the pathology of Paget-Schroetter syndrome in order to make the correct diagnosis. This is the



Fig. 9.8 Collateral circulation due to a thrombosis of the subclavian vein (Paget-Schroetter syndrome)

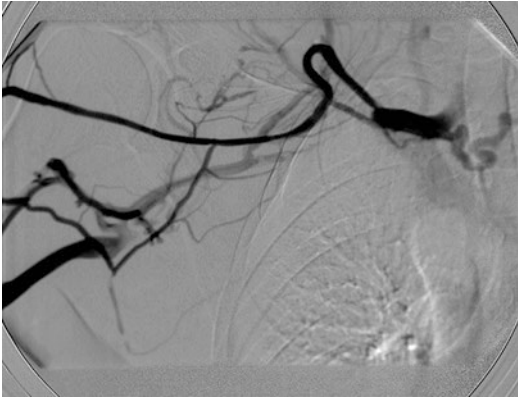


Fig. 9.9 Phlebography of a shoulder showing a thrombosis of the subclavian vein (Paget-Schroetter syndrome)

only way to prevent serious potential complications, such as pulmonary embolism. In addition to the clinical examination, color-coded Doppler sonography is the most important diagnostic tool. Phlebography (Fig. 9.9) (or MRI angiography) is only necessary in rare cases and if the findings are unclear. In addition, CT and MRI may provide important information about additional causes of compression (e.g., cervical rib). Furthermore, thrombophilia diagnosis should be carried out over the course of the disease. Therapy is determined by the extent of the findings and aims to achieve freedom from symptoms as quickly as possible and reduce the risk of developing a post-thrombotic syndrome.

In addition to conservative therapy using oral anticoagulation, surgical options for recanalization and even addressing anatomical variants (e.g., resection of a cervical rib) are available [30, 31].

Thrombosis of the Subclavian Vein (Paget-Schroetter Syndrome)

- **History:** Functional or anatomical compression of subclavian and axillary vein
- **Onset:** Acute to chronic
- **Clinical:** Pain, swelling, and cyanosis of the affected arm
- **Diagnostic:** Medical history, clinical finding, color-coded Doppler sonography, phlebography, or MRI angiography if applicable

- **Therapy:** Conservatively (oral anticoagulation), possibly surgically depending on the extent of the findings
- **Outcome:** Often long-lasting complaints

9.8 Conclusion

Shoulder injuries are the second most common pathologies in rock climbing after finger injuries. SLAP lesions (superior labrum anterior to posterior) play an especially essential role.

Furthermore, the frequency of pathologies of the rotator cuff, the long head of the biceps tendon, impingement symptoms, and injuries due to a shoulder dislocation has increased recently. The basic pillars of diagnosis are a thorough clinical investigation, sonography, and MRI. If surgical therapy is necessary, recent studies have shown very promising postoperative results, enabling rock climbers to resume their pre-injury climbing abilities.

Furthermore, targeted shoulder stability training (e.g., www.act.clinic) as well as avoiding passive dead hangs in the shoulder joint can help to prevent shoulder injuries [26]. Also closed chain eccentric training has recently proofed to help in prevention [32].

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